

Natural Phenomena Revision Pack

Observation, mechanism, myth and evidence

Review fifteen representative phenomena from light, atmosphere, water, sound, matter, electricity and space. Answers are separated at the end for independent practice.

Clear explanations - consistent SI units - ready to print

Light and atmosphere

Rainbow

Mechanism: Refraction, dispersion and internal reflection in water droplets.

Your observation, diagram or example:

Blue sky

Mechanism: Shorter visible wavelengths are scattered strongly by air molecules.

Your observation, diagram or example:

Mirage

Mechanism: Light curves through air layers with different temperatures and refractive indices.

Your observation, diagram or example:

Water and weather

Clouds

Mechanism: Tiny droplets or ice crystals have low settling speeds and are supported by air motion.

Your observation, diagram or example:

Dew

Mechanism: Water vapour condenses when a surface cools below the dew point.

Your observation, diagram or example:

Top-down freezing

Mechanism: Water is densest near 4 C; ice is less dense and floats.

Your observation, diagram or example:

Sound and electricity

Echo

Mechanism: Reflected sound returns after a measurable delay.

Your observation, diagram or example:

Balloon attraction

Mechanism: Transferred charge polarizes a nearby neutral surface.

Your observation, diagram or example:

Lightning and thunder

Mechanism: Light reaches us far sooner than the associated sound.

Your observation, diagram or example:

Knowledge check

1. Why is a rainbow not located at one fixed physical position?

2. Why does the Sun appear redder near the horizon?

3. Why can a lake remain liquid beneath surface ice?

4. Why does a rubbed balloon attract a neutral wall?

5. What information can be estimated from the delay between lightning and thunder?

6. State one condition that can change the boiling temperature of water.

Answers and explanations

1. The bow depends on the observer, Sun and a particular set of droplets at the required angles.
2. The longer atmospheric path scatters more shorter-wavelength light out of the direct beam.
3. Floating ice insulates the water, and the density behaviour of water limits overturning below about 4 C.
4. The electric field rearranges charge slightly in the wall, bringing opposite charge closer on average.
5. With an approximate sound speed, the delay provides an estimate of distance to the lightning.
6. Surrounding pressure, altitude and dissolved substances can change the boiling temperature.

Source guidance

For deeper review, use the cited NASA, NOAA, NIST, Institute of Physics and university sources linked on the corresponding EduVerse pages.